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eBloomDQG: Integrating Cognitive Level Alignment with Encoder-Decoder Architecture for Deep Question Generation

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ABSTRACT Creating exam questions that align with appropriate cognitive levels is a time-consuming task for teachers. It requires not only alignment with subject content but also the targeted cognitive level. To address this issue, Automatic Question Generation (AQG) has been developed to generate questions automatically from educational texts. However, most existing AQG systems focus on producing well-structured without alignment with cognitive level. Previous studies have improved certain aspects of question generation, such as language fluency, but do not integrate with cognitive levels. To address this gap, this study proposes a Deep Question Generation (DQG) framework that integrates a cognitive guide called eBloom, an extended version of Bloom's Taxonomy. eBloom combines educational theories to better represent different cognitive levels. The framework adopts an encoder-decoder architecture, where the encoder captures sequential and semantic information from the input text, while the decoder generates questions using a language model. The generation process is guided by eBloom to ensure alignment with the targeted cognitive level, while a knowledge base is incorporated to provide additional contextual information for question generation. The framework is evaluated on two datasets: a Social Studies dataset from Indonesian textbooks and the EduProbe dataset. Evaluation uses automatic metrics and expert assessment. Results show that the proposed model achieves a BERTScore of 72,59%, compared to 58,17% on baseline, with improvements of up to 25%. This indicates that combining proposed framework with eBloom can produce questions not only fluent and easy to understand but also aligned with the appropriate cognitive level.

INDEX TERMS Automatic question generation, bloom's taxonomy, deep question generation, encoder-decoder architecture, natural language processing

I. INTRODUCTION

IN education, formulating questions that align with both the subject content and targeted cognitive levels is a time-consuming task. Each question not only assesses students' understanding but also represents certain cognitive levels, thus it will affect its format and learning outcomes. Consequently, teachers must ensure this alignment by classifying each question based on Bloom's Taxonomy, a framework that categorizes cognitive abilities from basic remembering to higher order creation [1]. Despite widespread use, Bloom's Taxonomy has limitations particularly due to the overlap between cognitive levels. As highlighted by Das et al. [2], keyword based cognitive classification fails to accurately capture intended cognitive categories, for example words like describe, explain, and show corresponding to multiple levels depending on the context. To illustrate, the question *"explain the difference between absolute location and relative location!"* typically categorize as understanding level, while *"explain why economic inequality persists despite regional development policies!"* require deeper analysis. These examples highlight the same keyword corresponding to different cognitive levels when the context changes.

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The need for clarity in determining cognitive levels becomes increasingly important when question development is conducted automatically. An approach that enables to handle question generation from learning materials is known as Automatic Question Generation (AQG) [3]. AQG aims to automatically transform textual learning materials into questions that align with learning objectives [4]. In general, AQG approaches can be categorized into rule-based and neural network-based methods [5]. Rule-based approaches rely on predefined linguistic patterns and templates, which are can use for generating simple factual questions but struggle with complex sentence structures and contextual variation [6]. In contrast, neural network-based approaches learn patterns directly from data, enabling them to generate more diverse and context-aware questions, including reasoning and inference-based types [5].

The neural-network based methods in Automatic Question Generation (AQG) strongly depends on the architecture used. Early models such as RNN, LSTM, and GRU process text sequentially and can capture context to some extent [7]. However, they often struggle to understand long-range dependencies and more complex semantic relationships. The introduction of Transformer architectures, which use a self-attention mechanism, has significantly improved this capability [8]. Transformers can better capture relationships between words, even when they are far apart, allowing them to generate more coherent and contextually relevant questions. As a result, they consistently outperform RNN and LSTM-based models on benchmark datasets such as SQuAD and HotpotQA [4]. This outstanding performance is primarily attributed to the encoder-decoder architecture adopted by Transformer-based models, which enables more effective understanding and generation of contextually relevant questions. The encoder-decoder architecture has demonstrated remarkable versatility in capturing complex data representations across various computational domains. While it has been successfully adapted for unsupervised data clustering and robust matrix factorization tasks [9], our framework adopts this paradigm specifically for sequence-to-sequence natural language generation.

However, despite these methods ability to use language well, most existing AQG methods primarily do not explicitly incorporate do not match with learning goals [5]. As a result, while they may generate well-formed questions, they often lack alignment with learning goals, particularly in representing cognitive levels by Bloom's Taxonomy. Several studies highlight that relying solely on hierarchical cognitive levels is insufficient, as it does not fully capture the characteristics and intent of a question [10]. To overcome these limitations, it is necessary to complement Bloom's Taxonomy with additional information that provides more explicit semantic and contextual guidance for question construction. Therefore, integrating multiple educational theories can provide a more comprehensive representation of cognitive processes and question types [3].

One such theoretical framework is Graesser and Pearson's taxonomy [11], which refines question classification by focusing on the depth of inquiry. In parallel, Halliday's linguistic theory [12] explains how meaning is constructed through functional language choices, emphasizing the role of context, semantics, and grammatical structures. Furthermore, Dewey's inquiry model [13] provides a procedural view of learning by organizing thinking into sequential processes, such as identifying problems, formulating hypotheses, and drawing conclusions. Complementing this, Scott's framework [14] emphasizes problem-oriented and critical questioning, guiding the formulation of questions that encourage analysis, evaluation, and solution-based thinking in both open-ended and structured formats. By synthesizing these four frameworks with Bloom's Taxonomy, this study constructs an enriched cognitive framework referred to as eBloom, which extends beyond single-level classification into an integrated perspective that captures question intent, structure, linguistic features, and reasoning depth, thereby reducing ambiguity caused by overlapping cognitive levels.

It is important to note that while eBloom retains the six-level structure of Bloom's Taxonomy (C1-C6), the distinction lies not in the number of levels but in the depth of representation within each level. Unlike standard Bloom's Taxonomy, which primarily characterizes each level through cognitive action verbs, eBloom enriches each level using four complementary theoretical dimensions: question intent and inquiry depth (Graesser & Pearson), functional linguistic features (Halliday), sequential cognitive processes (Dewey), and problem-solving orientation (Scott). This multi-dimensional representation enables the six cognitive-level embeddings to capture substantially richer and more differentiated semantic information, thereby partially addressing the intra-class heterogeneity that would otherwise arise from a simple six-level classification.

Although eBloom provides a clear of cognitive levels, its teaching insights need to be changed into a form that can help guide automatic question generation. To bridge this gap, this study constructs a knowledge base using a Large Language Model (LLM), which organizes and encodes the cognitive level derived from eBloom that can be accessed during the question generation process. In this framework, eBloom is embedded within the knowledge base, which stores context, questions, and cognitive levels based on eBloom. Together, they form a fundamental of the proposed framework, ensuring that the generation process is guided not only by structured languange but also based on cognitive level.

Building on this foundation, this study proposes a Deep Question Generation (DQG) framework that brings together three core elements: an encoder-decoder architecture, eBloom as the cognitive guide, and a knowledge base as the contextual information. The encoder captures both the sequential flow and semantic relationships within the input text, while the knowledge base incorporates eBloom-derived cognitive representations into the generation process. The decoder then generates questions that are not only linguistically coherent and contextually relevant but also aligned with

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appropriate cognitive levels. The framework is evaluated on two datasets, namely a Social Studies dataset derived from Indonesian textbooks and the publicly available EduProbe dataset, using both automatic metrics and expert assessment to validate its effectiveness.

The main contribution of this study is developing a Deep Question Generation (DQG) that integrates an encoder-decoder architecture with an enriched knowledge base. The proposed system captures question characteristics and cognitive levels across educational framework, including categories, linguistic features, inquiry models, and problem solving orientation. This integration enables the generation of linguistically fluent, contextually relevance, and cognitively appropriate. The proposed framework is demonstrated through its ability to automatically produce well-structured questions.

The organization of this paper is as follows: Section II reviews the existing literature on AQG utilizing encoder decoder models and recent advances in neural sequence-to sequence modelling. Section III describes the methodology of this proposed method. Section IV presents the experimental setup and experiment scenario of this study. Section V reports the experimental findings, validating the system's performance through quantitative performance metrics and qualitative analysis by experts. Section VI concludes the paper by summarizing the key findings, discussing the limitations and suggesting possible directions for future research and improvements.

II. RELATED WORKS

The development of AQG has evolved substantially, transitioning from deterministic rule-based systems to adaptive methods powered by generative artificial intelligence. Table 1 presents a comparison of previous studies on AQG, highlighting differences in datasets, knowledge bases, model architecture, and main contributions. Early research, such as Wijanarko et al. [15], focused on rule-based systems using key-phrase extraction and context-free grammar to generate questions from software engineering course materials. This study aims to ensure that generated questions are aligned with specific learning outcomes and skills, although it did not incorporate any external knowledge base. More recent works have increasingly adopted LLM and semantic representations to enhance question quality and contextual understanding.

Based on these earlier works, recent studies have changed towards leveraging advanced neural architectures and richer semantic representations to improve question generation. For example, Guan et al. [4] proposed a Semantic-Rich Reinforcement Learning-Deep Question Generation (SRL-DQG) model, which combines reinforcement learning, where the model improves through trial and error, with semantic graphs processed by Gated Graph Neural Networks (GGNN). Semantic graphs represent the meaning and relationships between concepts, allowing the model to generate more contextually relevant and coherent questions. Evaluations were conducted on two datasets: SQuAD, a large-scale reading comprehension dataset with over 100,000 question-answer pairs from Wikipedia articles and HotpotQA, a multi-hop question answering dataset that requires reasoning across multiple supporting facts. Results showed that the proposed approach outperforms traditional models by better capturing complex contextual information across both datasets.

In the education domain, researchers have explored methods that integrate domain knowledge with LLM to enhance question generation. For example, Wang et al. [3] proposed a prompt-pattern AQG approach that incorporates a collective knowledge base derived from teachers. By embedding expert pedagogical knowledge into the prompt structure, the model can generate questions that are not only syntactically correct but also pedagogically meaningful, aligning with learning objectives. Building on this idea, another study, Al Faraby et al [16] evaluated the performance of LLM such as GPT-3.5 Turbo and LLaMA-2-13B in both classifying and generating questions. This evaluation used multiple prompting strategies and aggregated outputs through a voting mechanism. The questions generated were further assessed by human evaluators, including experts and crowdsourced participants, to ensure accuracy, relevance, and alignment with learning outcomes. These studies demonstrate that combining structured pedagogical knowledge with the generative capabilities of LLM can significantly improve the quality and educational value of automatically generated questions.

Other studies have focused on leveraging encoder-decoder architectures and advanced LLM to generate a wider variety of question types and to target different educational contexts. For instance, Goyal et al. [17] employed T5 model, which uses an encoder to process the input text and a decoder to generate output text. This model can produce diverse question types, including subjective questions, fill-in-the-blank items, yes/no questions, and multiple-choice questions, making it highly flexible for various assessment needs. The T5 model was trained and evaluated on widely used datasets such as SQuAD, QuAC, and BoolQ, which contain rich and varied question-answer pairs. By learning to transform input contexts directly into question-answer pairs, T5 demonstrates strong performance in generating syntactically correct and contextually relevant questions.

Meanwhile, research by Maity et al. [18] explored the use of state-of-the-art LLM, including GPT-4 Turbo and Gemini Pro, to automate question generation specifically from school-level textbook content. These models take advantage of LLMs' generative capabilities, allowing them to create educational questions that are aligned with learning objectives and subject-specific content. The study highlighted that using textbook contexts enables the generation of questions that are pedagogically meaningful and bridge the gap between general-purpose question generation and curriculum-aligned assessments. Evaluations in this work included both quantitative metrics and expert reviews, ensuring that the generated questions are accurate, contextually relevant, and educationally valuable.

Although previous studies have made significant advances, existing methods rarely combine knowledge bases with archi-

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TABLE 1. Gap analysis of our contributions against previous studies in terms of cognitive level classification, knowledge base integration, and model architecture for question generation

Study	Dataset	Knowledge Base	Model Architecture	Main Contributions
[15]	The Software Engineering course	None	Rule/Grammar-based (key-phrase extraction, context-free grammar dan question template)	Proposes a key-phrase-based question generation framework that ensures alignment with learning outcomes and skills.
[4]	SQuAD and HotpotQA	None	Encoder-Decoder and Reinforcement Learning (RL)	Propose an SRL-DQG model utilizing GGNN-based semantic graphs for question generation.
[3]	The Chinese language subject	Constructed by codifying teachers' pedagogical experiences	OpenAI (ChatGPT)	Introduces a prompt-pattern AQG method that integrates a collective teacher knowledge base with LLMs to generate higher-quality questions.
[16]	OpenStax.com	None	GPT-3.5 Turbo dan Llama-2-13B	Evaluates LLMs for classifying and generating educational questions using prompting techniques and human assessment.
[17]	SQuAD, QuAC and BoolQ	None	T5 (Text-to-Text Transfer Transformer)	Proposes an encoder-decoder T5 model to generate various question-answer types.
[18]	EduProbe Dataset	None	GPT-4 Turbo, GPT-3.5 Turbo, Llama-2-70B, Llama-3.1-405B, and Gemini Pro	Explores the use of LLMs to automate question generation from school-level textbook contexts.
Our Study	**Social Studies textbooks and EduProbe Dataset**	**Generated by LLM (Gemini-2.5-Pro) from textbooks**	**Encoder (sequential encoder and semantic graph encoder) and Decoder (GPT 2)**	**Proposes a DQG system that integrates encoder-decoder architecture with knowledge base.**

tectures that model both sequential and semantic relationships in text. This combination is important because sequential representations alone may miss conceptual connections between distant entities, while semantic representations alone may lack the linguistic coherence needed to produce grammatically well-formed questions. Capturing these semantic relationships is crucial for deep text comprehension. While some studies rely on word co-occurrence constraints to enrich semantic representations in unsupervised text clustering [19], our proposed Semantic Graph Encoder explicitly models the structural relationships of educational concepts to condition the generative process. This study takes a different approach by separating the role of LLM from the role of the generation model. Specifically, Gemini-2.5-Pro is employed not for direct question generation, but for constructing the Knowledge Base leveraging its strong reasoning capability to extract, structure, and encode complex cognitive knowledge derived from eBloom into machine-readable representations [20]. The actual question generation is then handled by GPT-2, as a decoder within a structured encoder-decoder architecture. GPT-2 is selected because it is open-source accessible, making it particularly suitable for research conducted under low-resource computational settings where deployment can be performed without requiring prohibitively expensive hardware. The encoder integrates two complementary approaches: a sequential encoder that captures the linear flow and syntactic patterns of the text, and a semantic graph encoder that models the relationships and meanings between concepts. By combining these elements, cognitive alignment is explicitly injected into the generation process through decoder process.

In summary, this study advances the field of educational question generation by proposing a framework that separates cognitive knowledge construction from question generation. Unlike prior studies that rely on a single model as an end-to-end generator, this study leverages Gemini-2.5-Pro specifically for building a structured knowledge base, while a GPT-2 decoder guided by combining sequential and semantic graph encoding ensures that the generated questions are not only linguistic fluent and contextually relevant, but also explicitly aligned with the intended cognitive level of eBloom. Here, cognitive level serves as a practical pedagogical guideline rather than an objective property inherent to every question; its application is context-dependent, shaped by instructional objectives and the learning goals of the assessment.

III. METHODOLOGY

This chapter describes the design, implementation, and evaluation of the DQG system built using an encoder–decoder architecture. The proposed system aims to generate questions with varying levels of cognition by integrating LLMs within the encoder–decoder architecture. Fig. 1 shows the general research workflow used in this study, while the detailed workflow of the proposed method can be seen in Fig. 1. As shown in the figure, the research process consists of three main stages: knowledge base creation, question generation, and performance evaluation. The architecture presented in Fig. 1 represents the primary contribution of this research, as it introduces a novel integration of sequential encoding and semantic graph encoding on the encoder side, with cognitive-level conditioning injected into the decoder to guide the question generation pipeline.

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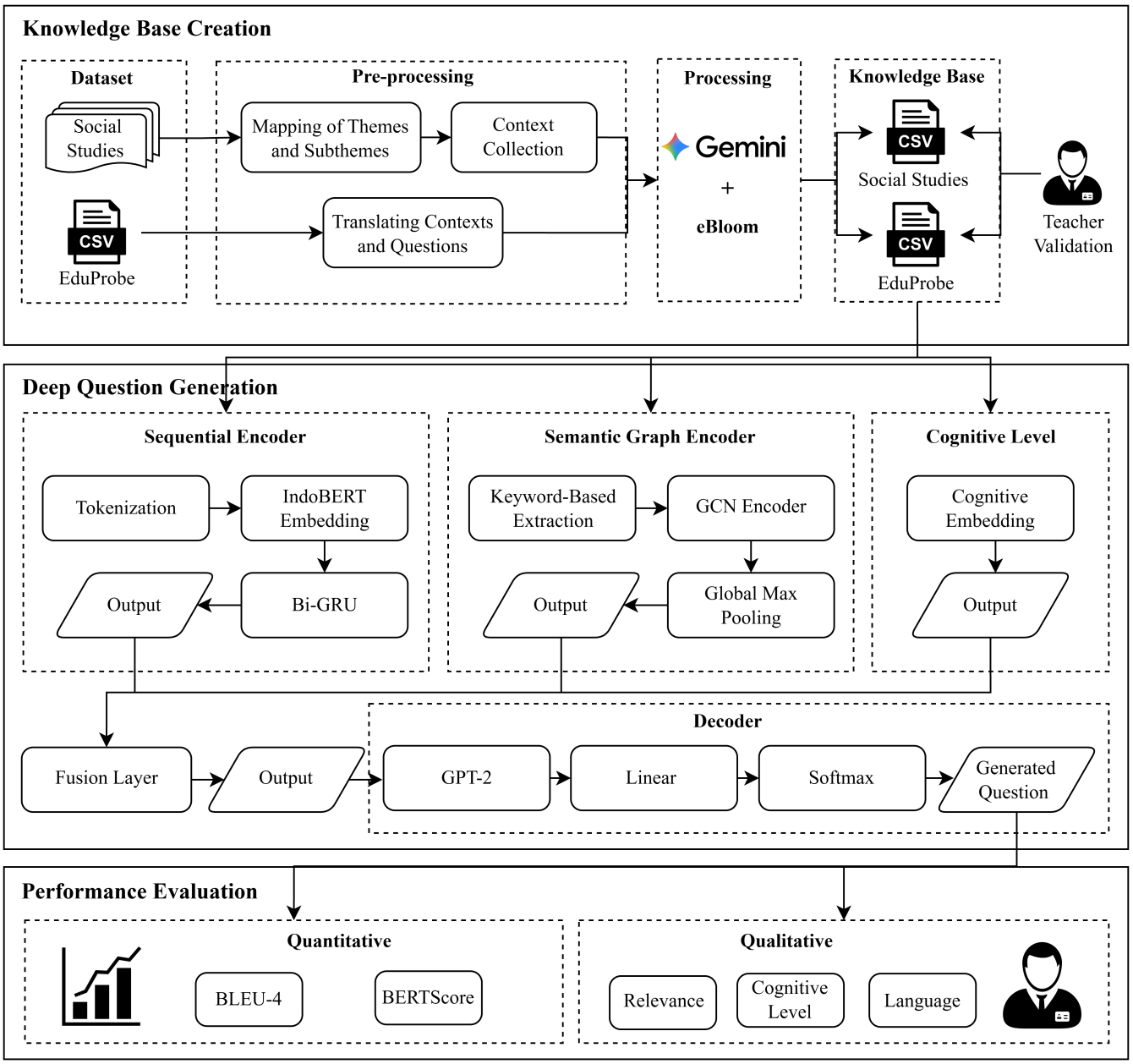


FIGURE 1. Proposed architecture of Deep Question Generation. The framework integrates IndoBERT-based sequential encoding, graph convolutional semantic encoding, gated attention fusion, and cognitive-level-conditioned GPT-2 decoding for automatic question generation, with experimental scenarios detailed in Table 8

A. KNOWLEDGE-BASE CREATION WITH EBLOOM

The application of Bloom's Taxonomy in automatic question generation and classification still has limitations, as the determination of cognitive levels often relies on keywords associated with each Bloom's Taxonomy level (see Table 2). These challenges where the same expression may correspond to different cognitive processes depending on context make consistent and reliable classification difficult, especially when explicit linguistic cues are absent.

To address ambiguity and overlapping categories, this study introduces an enriched cognitive framework, referred to as eBloom. The proposed framework extends Bloom's Taxonomy by integrating multiple educational theories, including Graesser and Pearson's Taxonomy, Halliday's Theory, Dewey's inquiry-based learning theory, and Scott's Theory (see Table 3). Through this integration, eBloom offers a more structured, context-aware representation that enables more precise and consistent identification of cognitive depth in questions, making it suitable for applications such as automatic question generation.

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TABLE 2. The example of linguistic ambiguity in Bloom's Taxonomy including action verbs, question words, sentence structure and no explicit markers

Linguistic Feature	Keywords	Example	The Reason of Overlapping
Action Verbs	Choose, Describe, Design, Explain, Show, Use	**C2 (Understanding):** Explain the definition of social mobility! **C4 (Analyzing):** Explain the impact of rapid urbanization on economic disparity!	The word **"Explain"** forces traditional Bloom to use a single label. However, explaining a textbook definition (C2) is cognitively different from explaining a complex cause and effect relationship (C4). The single keyword ignores the object of the sentence.
WH-Word	Why	**C2 (Understanding):** Why is Indonesia located in the ring of fire? (Fact retrieval) **C4 (Analyzing):** Why did the 1998 monetary crisis lead to massive social reforms? (Logical reasoning)	**"Why"** frequently maps to C2 when asking for explicit textual facts but change to C4 when demanding implicit logical reasoning. Standard keyword matching misses this contextual shift entirely.
WH-Word	How	**C2 (Understanding):** How does the process of social assimilation occur? **C6 (Creating):** How can we design a community program to overcome youth unemployment?	The word **"How"** has an almost uniform distribution across all cognitive levels. It lacks inherent cognitive weight and relies entirely on the surrounding context (the specific inquiry task), making keyword mapping ineffective.
Sentence Structure (Question Stem)	What	**C1 (Remembering):** What does the term 'inflation' mean? **C2 (Understanding):** What does the central bank do to control inflation?	Bloom's taxonomy fails to differentiate the linguistic function of auxiliary verbs (e.g., **"mean"** for definition vs. **"do"** for function) when the primary question word "What" is identical, blurring the line between C1 and C2.
No Explicit Markers	(Questions without verbs/WH-words)	**C5 (Evaluate):** Is the relocation of the national capital a beneficial move for Indonesia's long-term economic growth?	Without explicit action verbs or standard WH-words, traditional Bloom is completely **"blind"** to the epistemic nature of the question, failing to recognize the high-level evaluative judgment required.

Knowledge base creation begins with dataset collection (as detailed in Section IV-A). These datasets are then prepared through pre-processing stage, where each dataset is processed differently. For the Social Studies dataset, themes and subthemes are identified to organize the context, as detailed in Section IV-A. For the EduProbe dataset, both contexts and questions are translated into Indonesian using Google Translate via document upload, do manually to ensure consistency with the local dataset.

After pre-processing, both datasets are processed using LLM (Gemini-2.5-Pro) guided by eBloom. For Social Studies dataset, LLM receives two inputs: context and eBloom (see Table 3). Then we generate questions by prompting the model using the instruction: *"You are a teacher tasked with creating questions based on the provided (CONTEXT) and cognitive level according to (EBLOOM)"*. For EduProbe dataset, the LLM is only tasked with assigning cognitive level labels to the existing questions, using the prompt: *"You are a teacher tasked with labeling the available questions. Please assign a cognitive level label according to (EBLOOM)"*. For each dataset, a single prompt template was used consistently rather than multiple prompt variants. This design choice was deliberate: since the eBloom specification already provides detailed level-specific guidance including action verbs, question words, and four complementary theoretical dimensions this structured information was used to guide question generation instead of introducing variation through different prompt wording. For the Social Studies dataset, a single prompt was issued for each of the 177 subthemes, requesting five questions simultaneously for all six cognitive levels (C1-C6), yielding 30 questions per prompt execution and a total of $177 \times 6 \times 5 = 5,310$ generated questions. For the EduProbe dataset, the labeling prompt was applied individually to each of the 1,000 existing questions to obtain an initial cognitive level label prior to expert validation. All generation and labeling processes were performed using Gemini-2.5-Pro to balance output diversity and consistency across iterations.. The outputs of these processes are two knowledge bases, namely Social Studies and EduProbe, stored in CSV format. Each entry in knowledge base consists of three components: context, question, and cognitive level.

The same prompt template was used consistently across all iterations after being informally tested on a small number of subthemes prior to full-scale generation. No automated pre-filtering, such as duplicate detection or format checking, was applied at this stage, as teacher validation was intentionally adopted as the primary quality control mechanism. This approach was chosen to ensure that the final cognitive-level labels reflected human pedagogical judgment rather than automated rules or heuristics. Finally, the constructed knowledge base is validated by teachers to ensure that the assigned cognitive levels are consistent with the questions and aligned with educational objectives. To assess the reliability of the teacher evaluations, the level of agreement among the evaluators is measured using Fleiss' Kappa, a statistical measure for inter-rater reliability in categorical data [21]. The Fleiss' Kappa coefficient is calculated as follows:

$$\kappa = \frac{\hat{P} - \hat{P}_e}{1 - \hat{P}_e} \tag{1}$$

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where $\hat{P}$ represents the observed agreement among raters and $\hat{P}_e$ represents the expected agreement by chance. The resulting kappa value is interpreted based on standard agreement levels, ranging from no agreement to almost perfect agreement. This evaluation ensures that the assigned cognitive levels are reliable and consistent across different evaluators. In this study, three experienced teachers independently evaluated each generated question based on the appropriateness of the assigned cognitive level according to eBloom. The resulting Fleiss' Kappa value of 0.82 indicates almost perfect agreement among the three validators, demonstrating that the cognitive level assessments are applied consistently and that the knowledge base is of sufficient quality for use in the question generation process.

The confusion matrix in Fig. 2 further illustrates the pattern of label changes before and after teacher validation across both datasets. In the Social Studies dataset, most cognitive level labels were retained by the validators, with C1 (100%), C4 (99.5%), and C5 (100%) achieving near-perfect retention, reflecting the clear and distinctive linguistic characteristics of these levels. The only notable revision occurred at C3, where 21.0% of questions were reassigned to C2, indicating boundary ambiguity between Applying and Understanding. This misclassification occurs because both C2 and C3 frequently share similar surface-level linguistic cues, such as the use of ''how'' and ''explain'' as question words. However, C2 (Understanding) requires the student to restate or interpret information already present in the text, whereas C3 (Applying) demands the use of that information in a new or practical context. Without explicit contextual markers distinguishing these two intents, the LLM tends to assign C3 to questions that validators judge as more accurately reflecting C2-level thinking.

In contrast, the EduProbe dataset exhibited considerably greater label revision, with major changes observed at C2 (40.0%), C3 (52.7%), C4 (41.4%), and C5 (59.0%), where most revisions were directed toward C2 (Understanding). This pattern is primarily attributable to the multi-domain nature of EduProbe, which spans both Social Science and Natural Science content. Unlike the Social Studies dataset, which employs domain-specific action verbs and contextual cues tied to each cognitive level, EduProbe questions frequently use generic question words such as ''what'', ''how many'', and ''why'' across all cognitive levels without sufficient contextual differentiation. As a result, the LLM struggles to identify the intended cognitive depth, leading to systematic over-assignment of higher cognitive levels. Validators then consistently reassigned these questions downward to C2 (Understanding), reflecting a tendency to apply more conservative and context-sensitive judgment when explicit cognitive markers are absent.

To further validate the cognitive level definitions in eBloom, an inter-rater agreement analysis was conducted to compare the consistency of cognitive-level assignments using standard Bloom's Taxonomy and eBloom. The results indicate that eBloom achieved a higher Kappa ($\kappa = 0.82$) than standard Bloom's Taxonomy ($\kappa = 0.70$), suggesting that the enriched cognitive-level descriptors in eBloom provide clearer operational guidance and enable more consistent classification across raters. As an additional quality control measure, automated readability metrics namely the Gunning Fog Index and average word count were computed for each cognitive level based on the finalized knowledge base. The Gunning Fog Index is a readability metric that estimates how difficult a text is to understand based on two main factors: sentence length and the proportion of complex words (words with three or more syllables). It is calculated as

$$GF = 0.4 \times \left(\frac{W}{S} + 100 \times \frac{C}{W} \right) \tag{2}$$

where W is the number of words, S is the number of sentences, and C is the number of complex words. The resulting score approximates the number of years of formal education required to easily understand the text on first reading, with higher values indicating greater linguistic complexity. Scores of 6 or below indicate text that is very easy to read (elementary school level), scores between 7 and 12 correspond to fairly readable text typical of middle to high school level, scores between 13 and 16 indicate difficult text requiring undergraduate-level reading ability, and scores of 17 and above indicate very difficult text typically associated with college-graduate or post-graduate reading level. Overall, both average word count and linguistic complexity tend to increase from lower to higher cognitive levels. Notably, C3 exhibits the highest average word count (23.45 words), exceeding even the higher-order cognitive levels. A similar pattern is observed for the Gunning Fog Index, which increases from C1 (21.71) and then remains relatively stable across C2-C6, ranging from 23.57 to 27.40, with the highest value observed at C4 (27.40). These findings suggest that the relationship between cognitive demand and linguistic complexity is not strictly linear, indicating that higher-order cognitive levels do not always correspond to proportionally higher language complexity.

Then, the finalized knowledge base is divided into training, validation and test sets with an 80:10:10 split ratio, where the training set is used to learn patterns in question generation, validation set is used to monitor and refine model performance, and the test set is used to evaluate the final model performance.

B. DEEP QUESTION GENERATION USING ENCODER-DECODER

The proposed model consists of two main components: a dual encoder and a decoder. The encoder combines a Sequential Encoder and a Semantic Graph Encoder. The Sequential Encoder processes the input text (context from knowledge base) token by token, capturing context and the positional information of answer spans within the passage, while the Semantic Graph Encoder captures the relationships between concepts in the text. Then, the decoder combines the information from both encoders and generates the target question.

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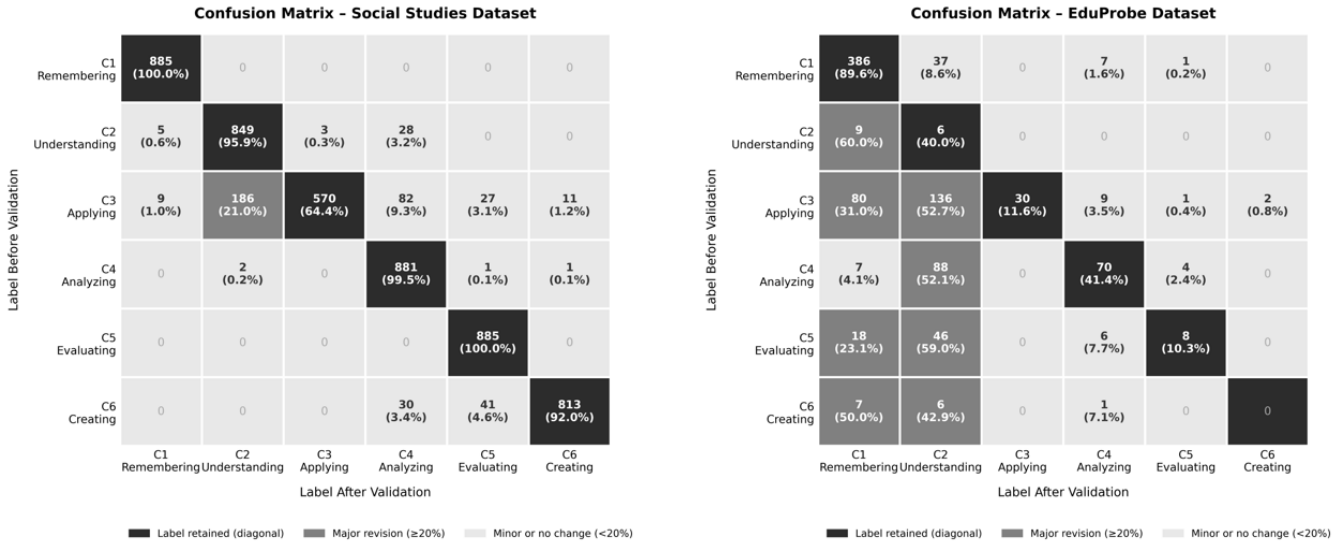


FIGURE 2. Confusion Matrix of Cognitive Level Labels Before and After Teacher Validation for the Social Studies Dataset (left) and EduProbe Dataset (right). Rows represent the original cognitive level labels assigned by the LLM prior to validation, and columns represent the final labels after teacher validation. Dark gray cells (diagonal) indicate labels that were retained by the validator, medium gray cells indicate major revisions ($\geq$ 20%), and light gray cells indicate minor or no change ($<$ 20%).

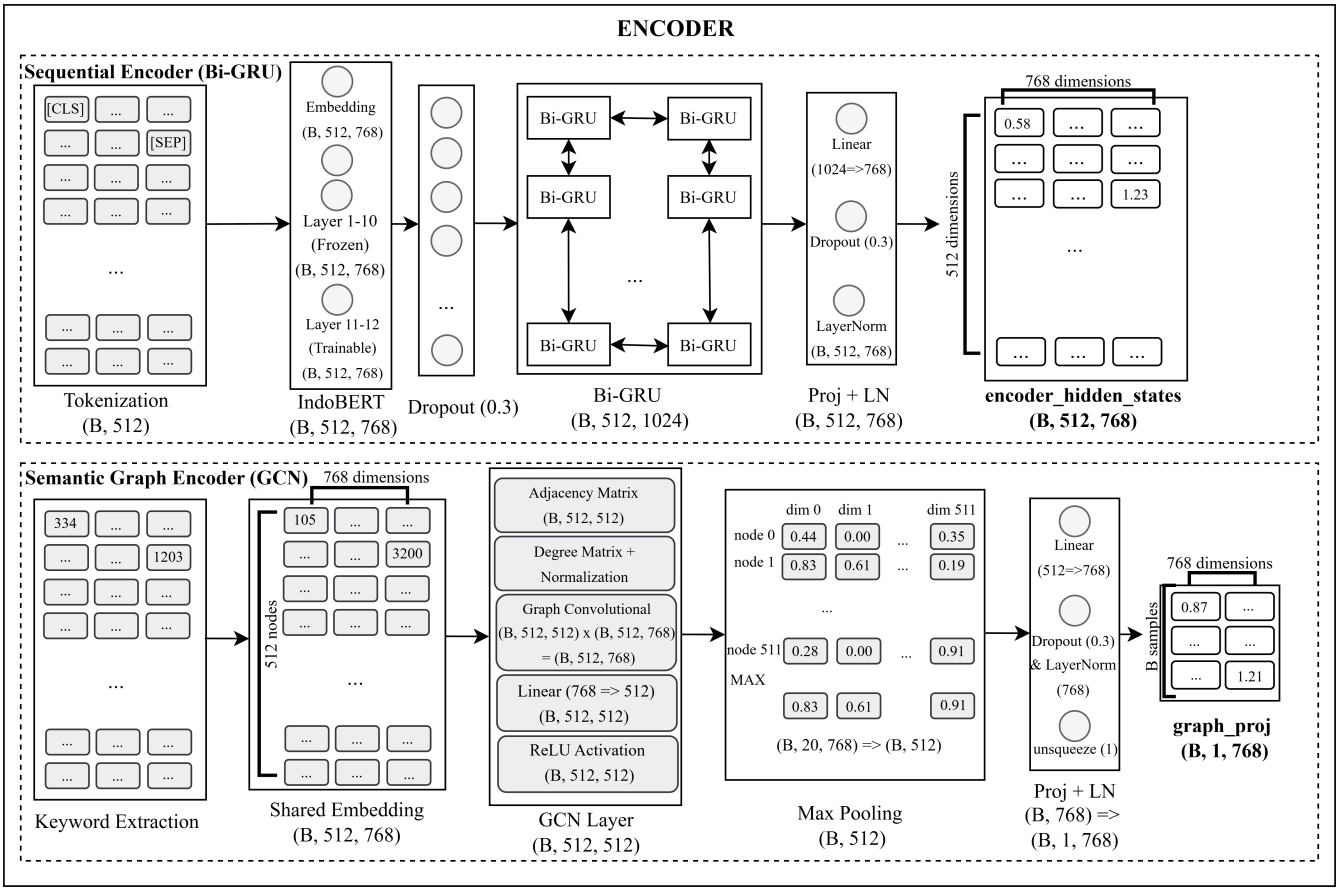


FIGURE 3. Architecture of the dual encoder module. (Top) The Sequential Encoder employs IndoBERT (layers 1-10 frozen, layers 11-12 trainable) followed by a Bi-GRU network. (Bottom) The Semantic Graph Encoder constructs a graph from extracted keywords using a shared embedding, processes it through a GCN layer.

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TABLE 3. eBloom mapping: an enriched cognitive classification system integrating bloom's taxonomy with multiple educational theories, including Graesser And Pearson's Taxonomy, Halliday's Linguistic Theory, John Dewey's Inquiry-Based Learning Theory, And Scott's Theory to provide a more structured and context-aware representation of cognitive depth in automatic question generation tasks

Bloom's Taxonomy	Linguistic Feature	Enriched Bloom (eBloom)	Example
C1-Remembering	**Action Verb:** list, name, state, mention, identify **Question Word:** who, when, where, how many, how much, is it true/false that...	**Graesser and Pearson's Taxonomy:** Verification Questions, Concept Questions, Quantification Questions and Disjunctive Questions **Scott's Theory:** Closed Questions	Mention one example in the text of the use of digital maps in daily life!
C2-Understanding	**Action Verb:** define, summarize, classify, categorize, paraphrase **Question Word:** what is the meaning/definition of..., what are the characteristic/features of..., explain the concept of...	**Graesser and Pearson's Taxonomy:** Definition Questions, Feature Specification Questions and Interpretation Questions **Halliday's Theory:** Classification Questions and Summary Questions **John Dewey's Theory:** Presenting the Problem **Scott's Theory:** Closed Questions	Explain the concept of the absolute location considered fixed and unchanging?
C3-Applying	**Action Verb:** apply, calculate, demonstrate, solve, execute **Question Word:** using x, how do you..., given the situation x, demonstrate...	**Graesser and Pearson's Taxonomy:** Procedural Questions **Scott's Theory:** Closed Questions	Considering that a new international airport is being constructed in a village, apply the concept of relative location to forecast the village's future development!
C4-Analyzing	**Action Verb:** analyze, compare, contrast, differentiate, corelate **Question Word:** why did x lead to y...?, what is the relationship/correlation between..., compare how x and y...	**Graesser and Pearson's Taxonomy:** Cause Questions, Effect Questions and Comparison Questions **Halliday's Theory:** Comparison Questions and Reasoning Questions **John Dewey's Theory:** Defining Questions and Drawing Conclusions **Scott's Theory:** Open-Ended Questions	Analyze how the expansion of railway networks and the adoption of Greenwich Mean Time (GMT) as a global time standard are interconnected!
C5-Evaluating	**Action Verb:** evaluate, assess, judge, justify, predict **Question Word:** to what extent do you agree with..., what would be the likely outcome/impact if..., is x the most effective way to... justify your answer	**Graesser and Pearson's Taxonomy:** Assessment Questions and Expectation Questions **Halliday's Theory:** Prediction Questions **John Dewey's Theory:** Hypothesis Verification **Scott's Theory:** Open-Ended Questions	Using information from the text, assess the benefits and drawbacks of online transport services on the social and economic well-being of the community!
C6-Creating	**Action Verb:** design, create, formulate, propose, invent, construct **Question Word:** how can we develop/design a plan to..., propose a new strategy/alternative for..., based on x, formulate a hypothesis about...	**Graesser and Pearson's Taxonomy:** Goal-Oriented Questions and Supporting Questions **John Dewey's Theory:** Formulating Hypotheses **Scott's Theory:** Open-Ended Questions	Make a digital map for your school event invitation. Using the text's examples as a guide, what features should you include to make it easier for invited visitors to reach the venue?

1) SEQUENTIAL ENCODER

The Sequential Encoder captures the syntactic structure and sequential dependencies of the input context. Initially, the context from the knowledge base is tokenized using a BERT-based tokenizer, which splits the text into subword units. Special tokens are inserted [CLS] at the beginning of the sequence and [SEP] at segment boundaries to preserve structural information. These tokenized sequences are then fed into a pre-trained IndoBERT model to generate dynamic, context-aware embeddings. The embeddings are subsequently processed through a Bidirectional Gated Recurrent Unit (Bi-GRU) to capture dependencies from both preceding and succeeding contexts. The output of the Bi-GRU is then passed through a linear projection layer followed by dropout and layer normalization to stabilize and regularize the representations, reducing overfitting while maintaining informative features. Finally, the processed features are projected into a fixed-dimensional encoder hidden state representation (768 dimensions), which serves as the contextual encoding of the sequential encoder process.

2) SEMANTIC GRAPH ENCODER

The Semantic Graph Encoder begins with a keyword extraction process, where key concepts are identified from the input context and represented as graph nodes. These extracted keywords are then mapped into a Shared Embedding matrix of shape (B, 512, 768), where each node is assigned a 768-dimensional vector representation across 512 nodes.

The embedded node representations are subsequently processed through the GCN Layer, which consists of several sequential operations. First, an Adjacency Matrix of shape (B, 512, 512) is constructed to represent the relationships between nodes. A Degree Matrix and Normalization are then applied to ensure balanced information propagation across the graph. The normalized adjacency matrix is passed through a Graph Convolutional operation of shape (B, 512, 512) $\times$ (B, 512, 768), producing an output of shape (B, 512, 768). This is followed by a Linear transformation that reduces the

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dimensionality from 768 to 512, resulting in a tensor of shape (B, 512, 512), and a ReLU Activation is applied to introduce non-linearity.

The output of the GCN Layer is then passed through a Max Pooling operation, which selects the most prominent features across all nodes, compressing the representation from (B, 512, 768) to (B, 512). This pooling operation ensures that only the most relevant structural information is retained for further processing. Finally, the pooled representation undergoes a Projection and Layer Normalization stage, where a Linear transformation expands the dimensionality back from 512 to 768, followed by Dropout (0.3) and LayerNorm to stabilize the representation. An unsqueeze operation is applied to add a batch dimension, producing a final graph projection output of shape (B, 1, 768), which serves as the semantic graph representation for integration with the sequential encoder in the fusion layer.

In terms of computational complexity, the semantic graph encoder can handle up to 512 nodes (512 tokens). Its computation mainly depends on the number of edges in the graph, rather than increasing sharply with sequence length. This makes it more efficient than a full self-attention mechanism with the same input size.

3) FUSION LAYER

The fusion layer aims to integrate the representations from the Sequential Encoder and Semantic Graph Encoder, into a single unified representation. This integration allows the model to capture sequential linguistic features and structural semantic relationships to generate questions that are contextually accurate. This concatenated representation is then passed through an Attention Gate, which produces a gating weight of shape (B, 512, 1) that dynamically determines the relevance of the structural. The gating weight is subsequently applied through a Gated Addition operation, enriching the sequential representation with the filtered external modalities while preserving the original linguistic flow, resulting in a tensor of shape (B, 512, 768). Finally, the fused representation undergoes a Final Concatenation to produce the Unified Encoder State of shape (B, 512, 768), which integrates contextual information, structural relationships, and cognitive instructions. This representation serves as the definitive global context for the decoder to generate questions that are linguistically fluent and contextually relevant.

4) COGNITIVE LEVEL

In the DQG process, the cognitive level serves as a guiding signal for the model to generate questions that align with the specified cognitive level. To incorporate cognitive level information into the model, cognitive level labels are encoded into dense vector representations. This allows the model to capture semantic relationships between cognitive levels and integrate them as additional features during prediction.

Specifically, the cognitive level input, represented as a batch of labels, is first mapped into an embedding matrix of size (6×768), where each of the six cognitive levels (C1–C6) is assigned a 768-dimensional vector. The resulting embeddings are then passed through a Layer Normalization (LayerNorm) to produce a normalized representation of shape ($B, 768$), which serves as the cognitive level feature for further processing. This embedding layer is implemented as `nn.Embedding(6, 768)` in PyTorch, where the weights are randomly initialized using a standard normal distribution and set as trainable parameters (`requires_grad=True` by default). No pretrained weights were used for initialization because the cognitive level labels (C1-C6) are task-specific categories with no direct equivalents in general-purpose pretrained language models. The embeddings are learned end-to-end during training via gradient-based optimization alongside all other model parameters.

5) DECODER PHASE

The decoder is responsible for generating questions word by word in an autoregressive manner, conditioned on both the encoded context representation and the specified cognitive level. By leveraging GPT-2 as the backbone, the decoder utilizes its pre-trained language generation capability to produce fluent and coherent questions. The decoder takes a sequence of input tokens consisting of special tokens [BOS], [C?], [QUES], and [EOS], represented as (B, seq, len). These tokens are then converted into dense vector representations through an embedding layer, producing a tensor of shape (B, seq, 768). The embedded representations are subsequently processed through the GPT-2 layers, which incorporate a Cross Attention mechanism of shape (B, seq, 768) to integrate contextual information from the encoder. The output is then passed through a Final Layer Normalization (Final LN), followed by a Language Model Head (LM Head) that projects the hidden states into vocabulary-sized logits of shape (B, seq, vocab). Finally, Beam Search is applied over the resulting token IDs to decode the most probable sequence, producing the Generated Question.

C. PERFORMANCE EVALUATION

The final stage of this research involves evaluating the quality of the questions generated by the system. This evaluation employs both quantitative and qualitative approaches. The quantitative evaluation measures the lexical and semantic similarity between the generated questions and reference questions, using BLEU and BERTScore metrics. These metrics provide an objective assessment of the model's performance in terms of language quality. Meanwhile, the qualitative evaluation focuses on the pedagogical quality of the generated questions, involving 5 expert evaluators to conduct the assessment. The evaluation considers several key aspects, including relevance, cognitive level, and language.

The Bilingual Evaluation Understudy (BLEU) metric works by calculating the similarity of word sequences (n-grams) between the generated text and the reference text (source). In this study, the metric is used to assess the quality of questions generated by the system. A high score indicates similarity between the word sequences in the generated text

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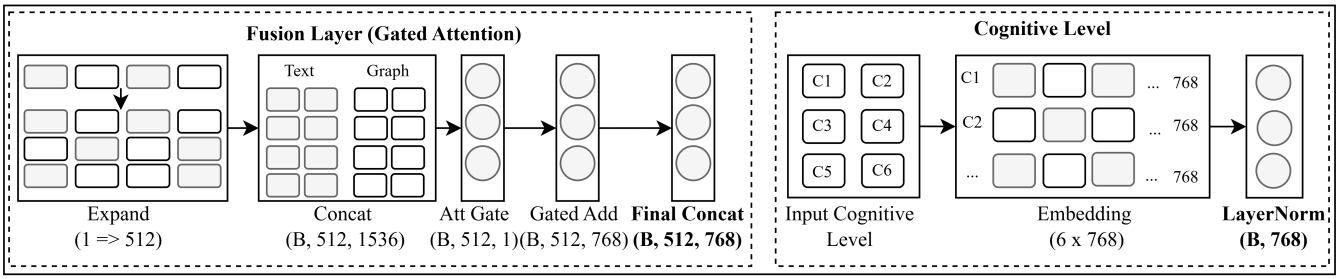


FIGURE 4. Architecture of the Fusion Layer and Cognitive Level Embedding modules. (Left) The Fusion Layer combines the Sequential Encoder and Semantic Graph Encoder outputs through concatenation, an Attention Gate, and a Gated Addition operation, producing a Final Concatenation of shape (B, 512, 768). (Right) The Cognitive Level module maps each of the six input cognitive levels (C1-C6) into a trainable 768-dimensional embedding vector via a learnable embedding matrix (6×768), followed by Layer Normalization to produce the final cognitive-level feature representation of shape (B, 768).

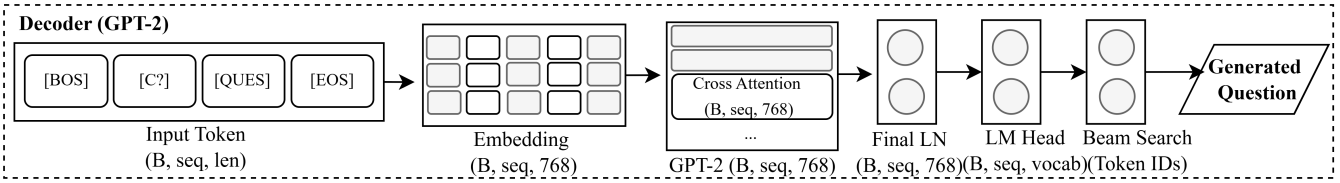


FIGURE 5. Decoder architecture based on GPT-2. The input token sequence ([BOS], [C?], [QUES], [EOS]) is passed through an embedding layer and processed by GPT-2 blocks with cross-attention, followed by a final layer normalization, a language model head, and beam search decoding to produce the generated question.

and the reference text (source) [22]. In addition, BERTScore measures the semantic similarity between generated questions and reference questions by computing cosine similarity over contextual embeddings from BERT. This metric ensures that the generated questions preserve the intended meaning and key information of the reference questions [23].

Relevance is used to measure how appropriate the questions are based on the context of the material being studied, as well as whether the answers to the compiled questions can be identified within the material [24]. The cognitive level evaluates the suitability between the intended cognitive level and the questions generated by the system, while also measuring students' understanding of a concept [25]. Meanwhile, the language aspect focuses on the clarity of the question sentences, ensuring that they are easy to understand, unambiguous, and can be answered directly, with language styles appropriate for level school students [18]. Each generated question will be evaluated using a Likert scale ranging from 1 to 5 is used in this study because it provides a clear and systematic way to collect expert judgments in numerical form [26]. Unlike binary assessments, the Likert scale allows evaluators to show different levels of quality, which is important when assessing detailed aspects of question quality such as clarity, cognitive alignment, and language suitability. This scale is commonly used in educational research because it is simple, reliable, and able to represent different expert opinions in a consistent and comparable way [27]. To ensure a clear and representative assessment, the evaluation results will be aggregated using the mean score provided by teachers

TABLE 4. Comparison of Question Distribution Across Educational Levels on The Experimental Data

Educational Level	EduProbe	Social Studies
Elementary School	303	-
Junior High School	395	1440
Senior High School	287	3870
University	15	-
Total Questions	**1,000**	**5,310**

TABLE 5. Distribution of Subjects and Number of Questions per Category Across Both Datasets

Category	Dataset: Subject (Number of Questions)
Social Sciences	**EduProbe:** History (221), Economics (174), Geography (240), Civics & Government (52), Social Issue & Culture (22) **Social Studies:** History (1,560), Sociology (1,140), Economics (1,350), Geography (1,260)
Natural Sciences	**EduProbe:** Biology (182), Chemistry (28), Physics (23)

[26]. This approach captures the overall judgment of the evaluators regarding the quality of the generated questions.

Combining the results of both quantitative and qualitative evaluations provides a comprehensive understanding of the DQG framework's ability to generate questions that are linguistically fluent, contextually relevance, and cognitively appropriate.

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TABLE 6. Hyperparameter details

No	Parameter	Value
1	ENC_EMBED_DIM	768
2	HIDDEN_DIM	512
3	DEC_EMBED_DIM	768
4	MAX_SEQ_LEN	512
5	MAX_NODES	512
6	BATCH_SIZE	8
7	NUM_EPOCHS	50
8	LEARNING_RATE	2e-5
9	PATIENCE	5
10	DROPOUT_RATE	0.3
11	WEIGHT_DECAY	0.01

IV. EXPERIMENTS

This chapter presents the experimental evaluation of the proposed framework, including system performance assessment and ablation analysis. The experiments are designed to assess the capability of the proposed model in generating cognitively aligned questions and its performance against established baseline architectures using standardized automatic evaluation metrics.

A. DATASETS

This study utilizes two datasets: (1) international public dataset, EduProbe [18] and (2) local dataset derived from Indonesian Social Studies textbooks. EduProbe[1] is a publicly available educational dataset consisting of 1,000 question-answer pairs spanning four educational levels (see Table 4). As shown in Table 5, the dataset covers subjects across two domains: Social Sciences and Natural Sciences. The dataset is provided in CSV format, consisting of two columns that store the context and its corresponding question.

The local dataset is compiled from Indonesian Social Studies textbooks at Grade VII-XII, obtained from the official Indonesian Book Information System[2] provided by the Ministry of Education. The dataset is distributed across Junior High School and Senior High School levels (Table 4). The textbooks are then systematically mapped into themes and subthemes to organize the content structure. Each subtheme is used to extract relevant content through an LLM process, where the model is prompted with the following instruction: *"You are a teacher, please summarize the material from subtheme 1-5"*. As shown in 5, it covers four subjects such as History, Sociology, Economics, and Geography, all within the Social Studies domain. This domain is selected due to its multidisciplinary nature and wide range of cognitive demands, from factual recall to critical analysis, making it well-suited for cognitive-level classification and question generation research.

B. EXPERIMENTAL SETUP

The experimental setup of the proposed model is configured based on a set of hyperparameters as summarized in Table 6. These parameters are determined to balance model performance and computational efficiency, as elaborated in the following discussion.

The proposed model is configured with IndoBERT as the tokenizer, with a maximum sequence length set to accommodate the full extent of educational contexts from the Social Studies dataset, which typically consist of multiple sentences, while remaining within the computational capacity of the IndoBERT architecture. The embedding dimension is aligned with IndoBERT's output representation to ensure compatibility between the tokenizer and the encoder components.

The sequential encoder employs a BiGRU with a hidden size that is doubled to capture both forward and backward contextual dependencies, while the semantic graph encoder produces an output dimension through the GCN layer that is kept consistent with the node size to maintain structural alignment. This alignment ensures that both representations share a compatible dimension during the fusion process. A dropout rate is applied across both encoder components to prevent overfitting and improve generalization. For the decoder, GPT-2 is selected due to its strong capability in generating natural language, having been pre-trained on a large and diverse text corpus that enables it to understand language structure effectively. Its autoregressive mechanism allows it to generate text step by step, making it suitable for constructing questions in a coherent sequence. To further improve the quality of the generated questions, Beam Search is employed as the decoding strategy, as it explores multiple possible output sequences simultaneously and selects the most coherent and probable result.

The training process is configured with a small batch size to accommodate GPU memory constraints while maintaining stable gradient updates. The number of epochs is set sufficiently large with an early stopping mechanism controlled by a patience parameter, allowing the model to converge without overfitting. A low learning rate is chosen to ensure stable fine-tuning of the pre-trained components, while weight decay is applied as an additional regularization technique to further prevent overfitting. The model supports six cognitive levels based on the eBloom taxonomy, enabling it to generate questions that are aligned with various levels of cognitive complexity.

To assess the computational feasibility of the proposed architecture, the inference time per batch was measured for each model configuration on an NVIDIA GeForce RTX 3060 (12 GB) GPU with a batch size of 8. The results show that M1 requires 1.84 seconds per batch, M4 requires 1.62 seconds per batch, the semantic graph encoder requires 1.67 seconds per batch, and the proposed M10 requires 2.20 seconds per batch. The per-question inference time of M10 is approximately 0.275 seconds, which remains within acceptable response times for practical deployment in educational settings.

C. EXPERIMENT AND ANALYSIS SCENARIO

To evaluate the proposed framework, we design a set of experimental scenarios that focus on comparative performance

[1] https://en.wikipedia.org/wiki/National_Council_of_Educational_Research_and_Training
[2] https://buku.kemendikdasmen.go.id/

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TABLE 7. Representative Samples of Contexts, Cognitive Level Classifications, Reference Questions, and Generated Questions from Both Datasets

Context	Cognitive Level	Reference Questions	Generated Questions
[Social Studies] History is the science that studies events, people, countries, or life that occurred in the past. Etymologically, 'history' comes from the Arabic word 'syajaratun' which means 'tree', symbolizing continuous growth and development. ...	C1	Sartono Kartodirdjo uses a multidimensional approach in his work entitled?	What is the definition of social science according to the text?
[Social Studies] Geography is the science that studies the interaction between humans and nature. Etymologically, geography comes from the Greek words 'geo' (earth) and 'graphien' (painting/description), so in short it means the science that describes the earth. ...	C2	Based on the context, explain why the regional approach is considered more comprehensive than the other two approaches!	Explain in your own words what is meant by bioregional and bottom-up!
[Social Studies] Sociology is a science that was born from the anxiety of experts regarding social changes and problems, especially those triggered by the industrial revolution and the French revolution in Europe in the 18th century. ...	C3	How can you apply the concept of socialization to teach your younger sibling about the importance of tolerance towards friends of different ethnicities?	You are a village head. How would you apply the principles of the bottom-up paradigm to address social problems in your village?
[Social Studies] The proclamation of Indonesian independence on August 17, 1945, was a historical milestone marking the nation's freedom from colonial rule. This event was heavily influenced by global political developments, particularly Japan's defeat in World War II. ...	C4	Identify the external (global) and internal (domestic) factors that were the main drivers of the Proclamation of Independence in August 1945!	Analyze the cause-effect relationship between Japan's defeat in Morotai and Balikpapan and the start of the Guerrilla War!
[Social Studies] The development of the modern trade system is driven by technological advances, particularly in transportation, which optimizes the movement of commodities, and telecommunications, which expands interactions between sellers and buyers. These advances have given rise to easy, fast, and efficient trade. ...	C5	Critique society's reliance on digital payment systems. What are the potential drawbacks or negative impacts for certain groups in society?	Which do you think has a greater impact on a country's economy? Justify your choice!
[Social Studies] The history of money begins in the pre-monetary era, where humans met their needs directly from nature through communal ownership. As humans began to settle, individual ownership and a barter system emerged, namely exchanging goods for other goods of equal value. ...	C6	Make an analogy to explain to your friend why paper money that is not backed by gold still has value!	Make a flowchart that describes the process of money entering Indonesia from abroad!
[EduProbe] According to the latest estimates in 2020, there are about 150 great Indian bustards left in India. Due to encroachment and poaching, their numbers continue to decline despite stringent conservation efforts.	C1	How many Great Indian Bustards are left in India?	How many Ganges River dolphins are left in the world?
[EduProbe] Human activities are also responsible for soil erosion. As the human population increases, the need for land also increases. Some human activities that cause soil erosion are: agricultural activities such as cutting down trees, plowing land to sow new seeds cause soil erosion. ...	C2	How do humans cause soil erosion?	What is meant by shifting cultivation?
[EduProbe] In the theory of continental drift, Wegener thought that all continents originally existed as a single supercontinent, Pangaea. He argued that about 200 million years ago, the supercontinent Pangaea began to break apart. ...	C3	How are plate tectonics related to continental drift?	How many states border the European continent and the European continent?
[EduProbe] The period from 1885 to 1905 is called the moderate period because they believed in the demands for reform while adopting constitutional and peaceful means to achieve their goals. ...	C4	Why is the period from 1885 to 1905 called the Moderate period?	What is meant by Deshmukh Act?
[EduProbe] One of the most important reasons to track air quality in a region is to monitor the condition of air and the state of pollution. ...	C5	What is the most important reason to track air quality in a region?	-
[EduProbe] Rajiv Gandhi was the Prime Minister who launched the Ganga Action Plan.Ganga Action Plan abbreviated as (GAP) is one of the longest and ambitious government interventions which have significantly influenced policies for controlling water pollution in India. ...	C6	Which Indian Prime Minister launched the Ganga Action Plan?	-

analysis across different model configurations.

a: Experiment 1: Dataset Characteristic Analysis

This experiment aims to evaluate whether the model consistently generates different types of questions for each cognitive level. The evaluation focuses on two main aspects of the generated questions.

First, the study examines the distribution of question keywords across each cognitive level to evaluate the model's ability based on cognitive demands. This is done by extracting the main question word from each generated question and measuring how frequently each keyword appears within a specific cognitive level. From this distribution, the degree of keyword overlap across multiple cognitive levels is then identified, revealing whether the model relies on a limited set of dominant keywords or employs a diverse and level-appropriate vocabulary.

Second, the length of the generated questions is analyzed by counting the number of words in each question. Descriptive statistics, including minimum, average, and maximum values, are calculated for each cognitive level. This analysis helps determine whether higher cognitive levels tend to pro-

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duce longer and more complex questions, while lower levels typically result in shorter and simpler ones.

b: Experiment 2: Ablation Study

An ablation study is conducted to evaluate the contribution of each component to the overall performance of the proposed model. A series of baseline models (M1-M9) are compared against the proposed model (M10), where each baseline represents a progressively more complex configuration with different combinations of components. For instance, M1 serves as the most basic baseline, utilizing a simple Encoder-Decoder architecture without any specialized components, while M3 introduces GPT-2 as both the encoder and decoder without incorporating the sequential or graph-based components. M4 extends this by adding the sequential encoder with GRU and cognitive level guidance, demonstrating the contribution of structured sequential processing. Each subsequent model incrementally adds or modifies components, allowing a systematic analysis of how each element contributes to the final performance. The full configuration of each model is detailed in the note of Table 8.

In addition to the automatic evaluation, a qualitative evaluation is also carried out to assess the quality of the generated questions from a human perspective. As shown in Table 8, human evaluators assess each generated question based on three aspects: whether the question is relevant to the given context, whether it matches the intended cognitive level, and whether it is grammatically correct and easy to understand. This evaluation aims to verify whether the performance improvements seen in Table 7 are also reflected in the actual quality of the generated questions.

Ultimately, the ablation study and the subsequent expert evaluations are designed to verify whether the proposed framework successfully achieves three primary objectives: producing questions that are linguistically fluent (measured by BLEU-4 and Language scores), contextually relevant (measured by BERTScore and Relevance scores), and cognitively appropriate (measured by the Cognitive Level expert assessment).

D. RESULTS OF EXPERIMENT 1: DATASET CHARACTERISTIC ANALYSIS

As presented in Fig. 6, keyword "what" appears at level C1 with a frequency of 5 and at level C2 with a frequency of 11 yet is also found at level C4 with a frequency of 2, indicating that the same question word is used across varying levels of cognitive demand. Furthermore, the keyword "how many" has the highest total frequency of 21, with the majority appearing at C1 level with 16 occurrences. However, it also appears at higher cognitive levels such as C3 and C4, which is unexpected because "how many" questions typically only require simple numerical answers and are more suitable for lower cognitive levels [29]. In contrast, the Social Studies dataset exhibits minimal overlap, where only the keyword "what" appears across more than one cognitive level, specifically at C1 and C2 with frequencies of 52 and 18, while all other keywords such as "analyze" and "create" remain exclusive to their cognitive levels.

This overlap pattern can be better understood by looking at how keywords are distributed within each dataset. As shown in Fig. 6, the keyword distribution in Social Studies is considerably more diverse, particularly at higher cognitive levels such as C4, C5, and C6, where each level is represented by specific action verbs. This occurs because higher cognitive levels demand more complex thinking operations, requiring more precise verbs to represent the corresponding cognitive actions [30]. In contrast, as presented in Fig. 6, EduProbe shows more overlapping keyword distribution across cognitive levels. This is because EduProbe does not use specific action verbs to distinguish between levels, meaning that the only distinguishing factor between cognitive levels lies in the content of the question itself rather than in the keywords used. As a result, the model finds it harder to learn the differences between cognitive levels from keyword patterns [2].

This overlap and distribution keywords are further supported by the length of the questions generated by the model at each cognitive level. As shown in Fig. 7, C5 exhibits the highest average number of words per question while C1 produces the shortest, suggesting that higher cognitive levels tend to require longer questions as the thinking process becomes more complex and demands more elaborate linguistic expression [30]. This is evident from the pattern where lower levels such as C1 and C2 produce shorter questions since they only require retrieval of basic information, whereas higher levels such as C3 and above produce longer questions as they require the inclusion of situational context or specific scenarios to properly guide the thinking process.

Based on the analysis of both graphs, keyword overlap and question length do not always have a positive relationship with BERTScore. In the EduProbe dataset, question words such as what, how many, and how appear across different Bloom's cognitive levels, but their BERTScore values remain low (0.181-0.216). This may happen because BERTScore measures similarity based on the meaning of the whole sentence, not just the question word. So, even if the same question word is used, the rest of the question can have a very different meaning from the reference. In contrast, the keyword what in the Social Studies dataset produces a higher BERTScore of 0.300, possibly because the topic is more focused, making the generated questions closer in meaning to the reference.

We clarify that the overall BERTScore of 76.67% reported for EduProbe represents the average performance across all subjects in the dataset, including Social Sciences, Natural Sciences, and other subjects. Because these subjects differ in textual characteristics, semantic matching may be easier for some than for others. When the comparison is limited to the Social Science subset of EduProbe, which is most similar in domain to the Social Studies dataset, the BERTScore decreases to 72.47%, nearly identical to the Social Studies score of 72.59%. This indicates that the model performs consistently across both datasets when they belong to the same domain, and that the higher overall EduProbe score is more

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TABLE 8. Comparison of experimental results of all models across all scenarios in the question generation task.

Models	Encoder			Decoder		Social Studies		EduProbe		EduProbe (SS)	
	Seq	Sem	LLM	Cog	LLM	B-4	BS	B-4	BS	B-4	BS
M1	-	-	N (Seq2Seq)	-	N (Seq2Seq)	0.0005	0.5817	0.0019	0.3699	0.0001	0.2049
M2	-	-	Y (GPT-2)	-	N (Seq2Seq)	0.0018	0.6017	0.0144	0.4514	0.0011	0.5312
M3	-	-	Y (GPT-2)	✓	Y (GPT-2)	0.0161	0.6908	**0.0526**	**0.7750**	0.0126	0.6897
M4	✓	-	N (GRU)	✓	N (GRU)	0.0412	0.7109	0.0328	0.7572	0.0338	0.3933
M5	✓	-	N (LSTM)	✓	N (LSTM)	0.0313	0.7156	0.0293	0.4128	0.0283	0.3930
M6 [28]	-	✓	N (GCN)	-	N (GRU)	0.0248	0.5208	0.0060	0.5486	0.0037	0.5142
M7	-	✓	N (GCN)	✓	N (GRU)	0.0109	0.6918	0.0194	0.7456	0.0083	0.6901
M8 [4]	✓	✓	N (GRU+GCN)	-	N (GRU)	0.0121	0.6499	0.0257	0.5861	0.0154	0.6667
M9	✓	✓	N (GRU+GCN)	-	Y (GPT-2)	0.0179	0.7007	0.0393	0.6273	0.0184	0.6974
M10	✓	✓	**N (GRU+GCN)**	✓	**Y (GPT-2)**	**0.0414**	**0.7259**	0.0398	0.7667	**0.0397**	**0.7247**

Note: Seq=Sequential Encoder; Sem=Semantic Graph Encoder; Cog=Cognitive Level; B-4=BLEU-4; BS=BERTScore; M1=Baseline Encoder-Decoder (Seq2Seq); M2=GPT-2 Encoder with Seq2Seq Decoder; M3=GPT-2 for both stages; M4=GRU Sequential Encoder; M5=LSTM Sequential Encoder; M6=Semantic Graph Encoder (GCN); M7=M6 with cognitive level; M8=Hybrid Sequential-Semantic Encoder; M9=Hybrid Encoder with GPT-2 Decoder; M10=Proposed complete framework.

TABLE 9. Comparison of qualitative evaluation results by teachers on architectural models and datasets across four aspects (Relevance, Answer Quality, Cognitive, and Language) using Social Studies and EduProbe

Models	Social Studies (%)			EduProbe (%)			EduProbe (Social Science) (%)		
	Rel	Cog	Lang	Rel	Cog	Lang	Rel	Cog	Lang
M1	82,0	89,3	79,0	55,3	67,0	58,6	53,3	55,3	63,3
M3	79,3	81,3	79,3	**60,0**	**60,0**	**75,3**	**56,6**	**60,0**	**70,0**
M4	74,6	87,3	86,6	50,6	54,0	72,0	50,0	53,3	70,0
M7	74,6	87,3	86,6	53,3	62,6	73,3	50,0	63,3	73,3
M10	**90,0**	**90,6**	**90,0**	54,6	57,3	71,9	46,6	56,6	**70,0**

Note: Rel = Relevance; Cog = Cognitive Level; Lang = Language

likely attributable to differences in subject mix rather than an actual performance advantage. We further clarify the distinction between keyword overlap and BERTScore, as these two measures capture different aspects of question quality. Keyword overlap reflects lexical similarity in question phrasing across cognitive levels, whereas BERTScore evaluates semantic similarity to the reference at the embedding level. As shown in Fig. 6, EduProbe exhibits high keyword overlap, where generic question words such as ''how many,'' ''what,'' and ''how'' recur across cognitive levels without specific differentiation, for instance, ''How many Great Indian Bustards are left in India?'' (C1) and ''How many states border the European continent?'' (C3) share the same interrogative structure despite targeting different cognitive levels. These shared stems cause generated questions to closely match reference phrasing at the embedding level, which consistently raises BERTScore across levels. In contrast, the Social Studies dataset uses level-specific action verbs that tie each cognitive level to a unique syntactic and semantic structure, increasing embedding distance from the reference even when the generated question is well-aligned to its cognitive level, resulting in a lower BERTScore despite better differentiation across cognitive levels.

In addition to wording variation, a structural difference in sentence formation is observed across cognitive levels. In Social Studies, questions at C1-C2 are typically posed as direct inquiries asking for a fact, definition, or basic explanation. Starting from C3, questions consistently shift toward embedded contextual or scenario-based framing, such as ''How can you apply the concept of socialization to teach your younger sibling about the importance of tolerance towards friends of different ethnicities?'' This contextual complexity continues from C4 onward through multi-condition, analytically demanding phrasing, such as identifying both global and domestic factors behind a historical event (C4) or constructing an analogy to explain an economic concept

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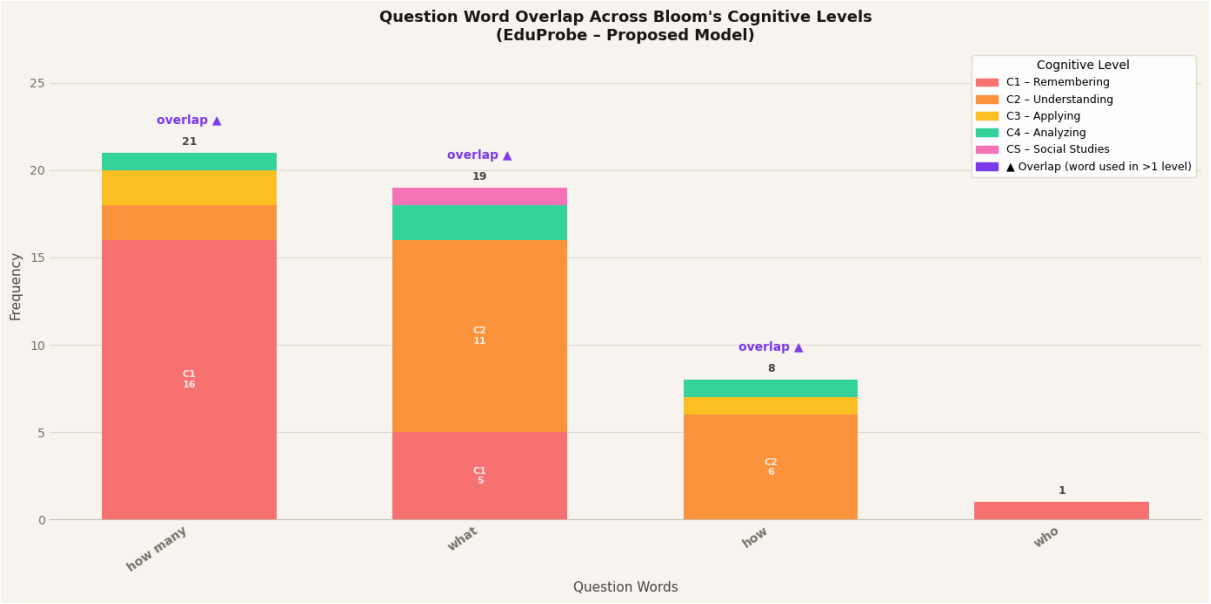


FIGURE 6. Keyword overlap across cognitive levels, showing the overlapping in every question word per question.

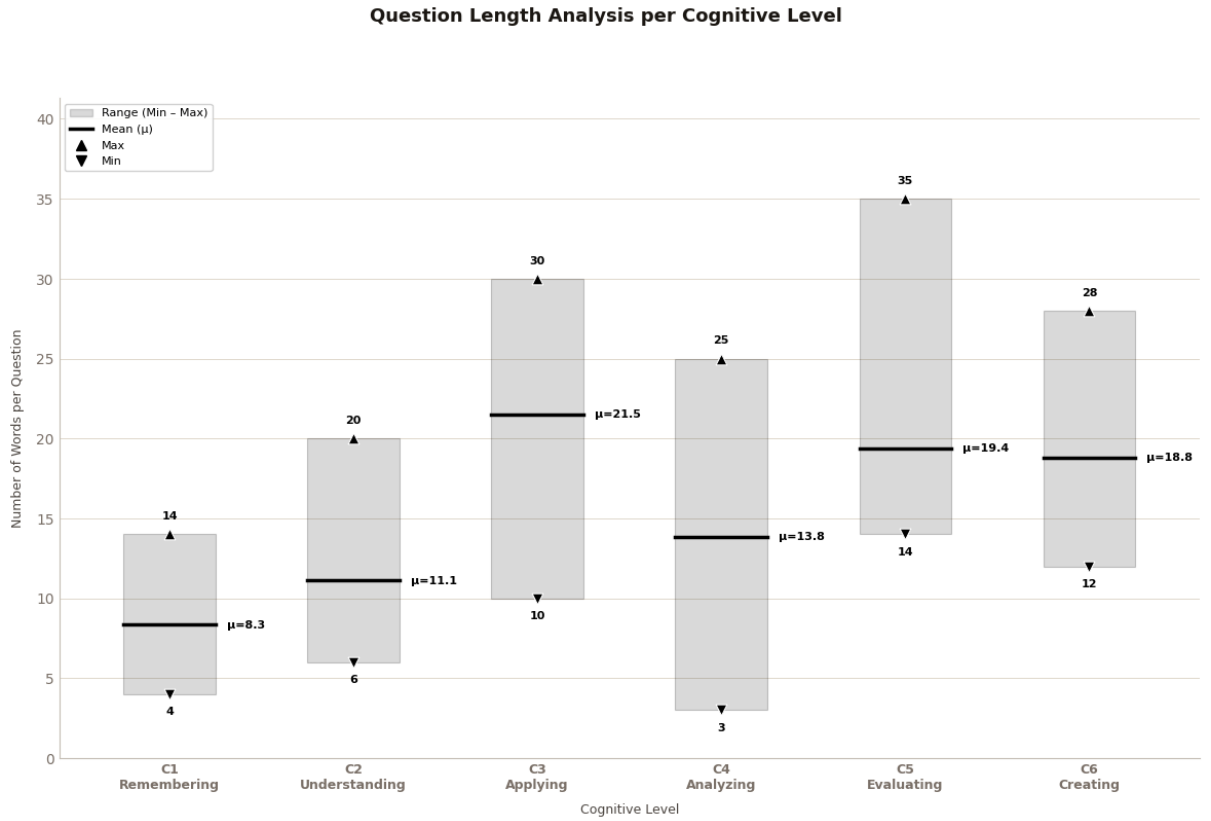


FIGURE 7. Distribution of question length across cognitive levels (C1-C6), showing the minimum, maximum, and mean number of words per question.

to a peer (C6), reinforcing the level-specific action verbs discussed above. In contrast, EduProbe does not consistently maintain this structural separation between lower- and higher-order questions. Reference questions such as ‘‘What is the most important reason to track air quality in a region?’’ (C5) and ‘‘Which Indian Prime Minister launched the Ganga Action Plan?’’ (C6) remain framed as direct factual inquiries even at the highest cognitive levels, without the contextual elaboration seen in Social Studies from C3 onward. This inconsistency in structural boundaries corresponds with the

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significant keyword overlap shown in Fig. 6,and clarifies why EduProbe questions at various levels are more similar to each other and to the reference at the embedding level, resulting in a higher BERTScore despite weaker cognitive-level differentiation. Table 8 presents representative reference questions from each dataset to illustrate this pattern. This interpretation is further supported by the qualitative results in Table 9, where M10 achieves the highest cognitive-level alignment score on Social Studies (90.6%), demonstrating that stronger cognitive-level differentiation does not necessarily correspond to a higher BERTScore. Taken together, these findings suggest that high keyword overlap does not directly lower BERTScore; rather, it primarily reflects and reinforces the absence of structural escalation across cognitive levels, a distinction better captured through qualitative cognitive-level evaluation than through BERTScore alone.

For question length, the Social Studies dataset shows that the longest questions (about 18.4 words on average) have a medium BERTScore of around 0.4, while shorter questions tend to get higher scores. This suggests that shorter questions are more general and easier to match in meaning with reference, while longer questions introduce more different words and ideas, which can lower similarity. Overall, these results show that the quality of generated questions is not only determined by question words or sentence length, but more by how well the overall meaning matches the reference questions.

Overall, the analysis of keyword overlap, BERTScore, and question length shows the same conclusion: the quality and complexity of generated questions depend more on the cognitive level they target than on surface features like question words or sentence length alone. Datasets with more varied and level-specific keywords, such as Social Studies, tend to produce questions that are closer in meaning to human references and have more appropriate lengths across different cognitive levels. In contrast, EduProbe uses more general question words across different levels, which makes it harder for the model to distinguish the level of thinking required. This results in lower BERTScore values and less consistent question lengths. These findings suggest that using level-specific words and clearer context in the generation process is important to produce questions that better match the intended cognitive level.

E. RESULTS OF EXPERIMENT 2: ABLATION STUDY

The following discussion examines the contribution of each architectural component to the overall performance of the proposed model, based on the ablation study results presented in Table 7.

As shown in Table 8, models that only use a sequential encoder, such as M4 and M5, perform 2%-19% better than models that only use a semantic graph encoder, such as M6 and M7. The sequential encoder based on GRU or LSTM processes words in order from left to right, allowing it to capture the grammatical structure and logical flow of a sentence, which is essential for generating questions that are well-structured and easy to understand. Without this structural foundation, the GCN-based semantic graph encoder tends to produce questions that are topically relevant but poorly structured, making it harder for the intended meaning to be clearly delivered [31]. However, as further shown in Table 8, combining both encoders as in M8, M9, and M10 improves performance by up to 14%, allowing the model to generate questions that are not only grammatically correct but also more accurately reflect the core meaning of the source text.

As shown in Table 8, M10 consistently outperforms M9 across all datasets, with performance improvements ranging from 1% to 3% across both metrics, showing that the cognitive level plays an important role in helping the model generate questions. In Table 7,without the cognitive level, M9 tends to produce unfocused questions such as *"Compare and contrast the factors that influence population growth in a country. Which has a greater influence on developing countries? Give your reasons!"*, which combines multiple sub-questions at once and makes it unclear whether the question targets C4 or C5. In contrast, M10 generates more precise questions such as *"Evaluate whether Contractionary Fiscal Policy (increasing government spending) is sufficient to address an economic recession?"*, which has a single clear focus, a specific cognitive action, and sufficient context to guide the thinking process without ambiguity. This confirms that the cognitive level functions as a guiding signal that ensures the generated questions are not only well-written and topically relevant, but also clearly matched to the intended cognitive level [32].

As shown in Table 7, using GPT-2 as the decoder consistently produces better results compared to conventional decoders such as GRU or LSTM. This is because GPT-2 has been trained in a very large and diverse collection of text, giving it a stronger ability to generate natural and fluent language. When GPT-2 receives representations enriched by the cognitive encoder, it is able to generate questions that are not only grammatically correct and meaningful, but also naturally written and easy to understand [33].

The quantitative improvements observed in Table 8 are further supported by the qualitative evaluation results in Table 9. As shown in Table 9, M10 achieves the highest scores on the Social Studies dataset across all three aspects, namely relevance, cognitive level, and language. This shows that M10 not only performs better in automatic metrics, but also produces questions that are relevant to the context, matched to the intended cognitive level, and grammatically correct. Compared to M1 as the baseline, M10 shows consistent improvement across all aspects and all datasets, indicating that every component added to the proposed model contributes to better question quality. However, on the EduProbe dataset, M10 does not always achieve the highest scores across all aspects. This is likely because EduProbe has limited variety in its keywords, making it harder for the model to differentiate between cognitive levels [34]. Overall, these results confirm that M10 is the best-performing model, as it consistently generates questions that are fluent, relevant, and clearly aligned with the intended cognitive level across different datasets.

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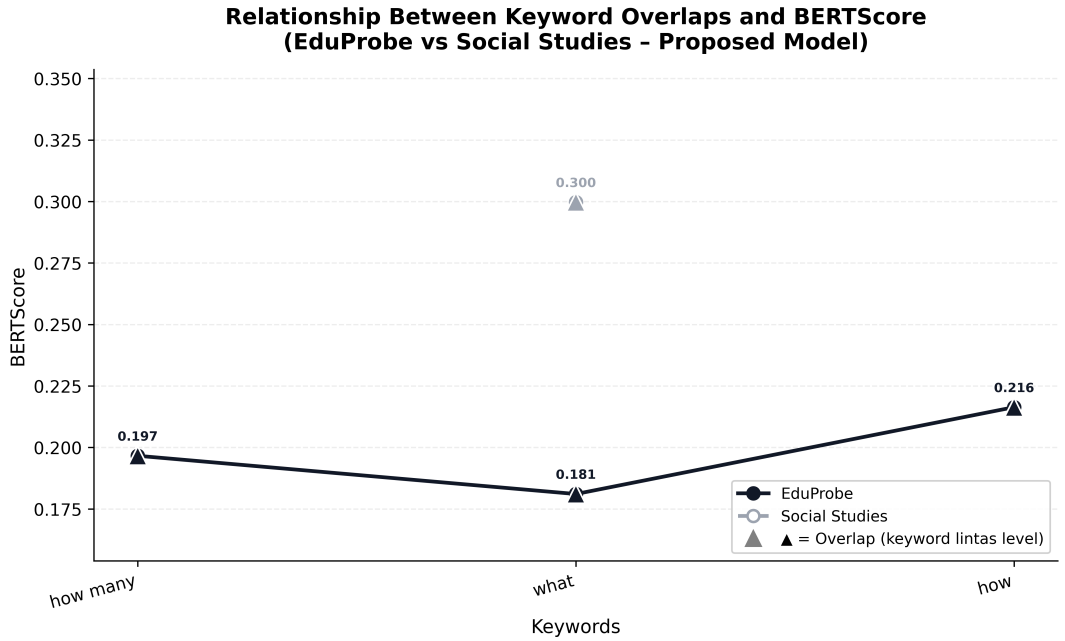


FIGURE 8. Relationships between keyword overlaps and BERTScore.

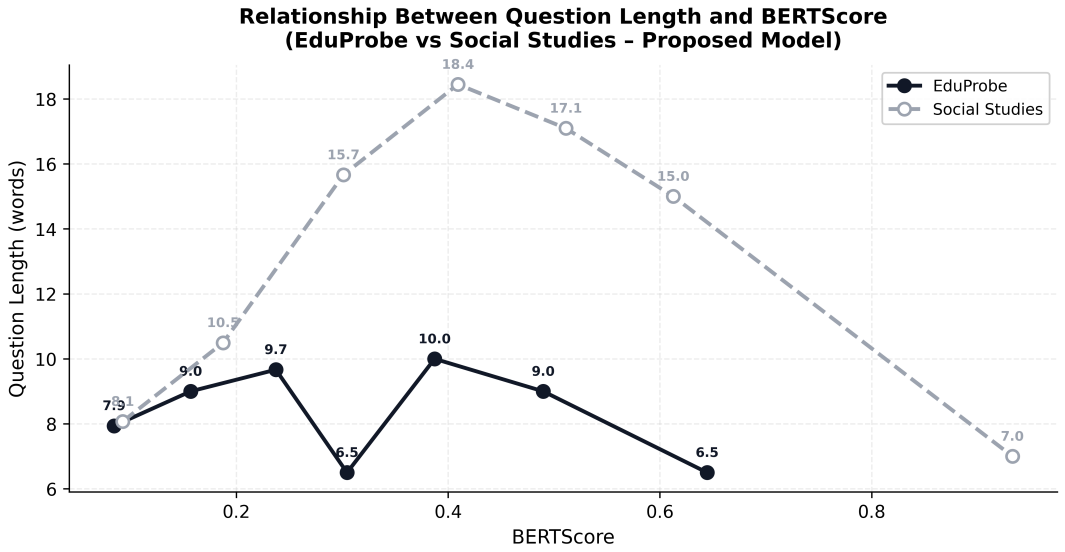


FIGURE 9. Relationships between question length and BERTScore.

The quantitative improvements observed in Table 8 directly influence the enhanced pedagogical quality reflected in the qualitative evaluation in Table 9. There is a clear positive correlation between the model's structural and semantic capabilities indicated by higher BLEU-4 and BERTScore metrics and the human expert assessments. Specifically, the ability of model M10 to accurately capture syntactic and semantic dependencies (as seen in its superior BERTScore of 72.59% for Social Studies in Table 8) provides the foundation for its highest scores in the Language and Relevance aspects in Table 9. Furthermore, the explicit inclusion of cognitive level guidance, which improved M10's computational performance over M9, is directly validated by the highest Cognitive Level rating (90.6%) from the teachers. This confirms that the architectural enhancements not only improve automated lexical matching but also effectively produce educational questions that are contextually relevant, grammatically clear, and cognitively aligned.

Overall, these results confirm that M10 is the best-performing model. The integration of the sequential encoder and GPT-2 decoder ensures the questions are linguistically fluent, while the semantic graph encoder guarantees they

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remain contextually relevant to the source text. Most importantly, the explicit cognitive level guidance allows M10 to consistently generate questions that are cognitively appropriate, outperforming baselines that lack this structured pedagogical control. To isolate the effect of the eBloom enrichment from other architectural factors, we further compared two models sharing an identical dual-encoder (sequential and semantic graph) architecture with a GPT-2 decoder, differing only in their cognitive-level representation: one used standard Bloom taxonomy labels, while the other used eBloom-enriched embeddings (the proposed M10). This controlled comparison was conducted on the Social Studies dataset. The eBloom model consistently outperformed the plain Bloom taxonomy model across all four evaluation metrics, achieving a BLEU-4 of 0.0397 compared to 0.0174, a ROUGE-L of 0.1585 compared to 0.1153, a METEOR of 0.1572 compared to 0.1193, and a BERTScore of 0.7247 compared to 0.7119. Since both models used the same architecture and all other components were kept unchanged, this improvement can be attributed specifically to the enriched cognitive-level representation rather than to differences in model architecture. These results support our claim that although eBloom retains the six-level structure of Bloom's Taxonomy, the richer information embedded within each level meaningfully reduces the ambiguity that a plain six-level representation alone would still carry.

V. DISCUSSION

A. KEY FINDINGS

This study presents a cognitive-level guided question generation model that integrates sequential, semantic, and cognitive components to produce questions that are linguistically fluent, contextually relevant, and cognitively appropriate. The following key findings are drawn from the experimental results.

The analysis of keyword overlap reveals that the ability of the model to differentiate questions across cognitive levels is heavily influenced by the linguistic characteristics of the dataset. As observed in the EduProbe dataset, the use of generic question words such as "what", "how", and "how many" across multiple cognitive levels results in a higher degree of overlap, making it harder for the model to learn clear distinctions between levels. In contrast, the Social Studies dataset, which employs both question words and specific action verbs for each cognitive level, provides clearer cognitive signals that help the model generate more level-appropriate questions. This finding highlight that the richness of cognitive level annotation directly affects the model's ability to produce differentiated questions across levels [35].

It is also important to note that the qualitative scores of 46.6-57.3% reported for the EduProbe Social Science subset are lower than the overall EduProbe qualitative scores (54.6-71.9%), as this subset contains fewer questions and shorter passages, which further limits the contextual information available for generating cognitively aligned questions. Furthermore, both models were trained and evaluated independently on their respective datasets; therefore, the performance gap between Social Studies and EduProbe does not indicate overfitting to the narrative style of Social Studies, but rather reflects inherent differences in dataset characteristics, including text length, subject diversity, and keyword variety.

The ablation study further demonstrates that combining the sequential and semantic graph encoders produces a stronger representation compared to using either encoder alone. The sequential encoder captures the grammatical structure and logical flow of the text, while the semantic graph encoder enriches the representation with conceptual relationships between entities. When combined, the two encoders complement each other, allowing the model to generate questions that are not only grammatically correct but also semantically meaningful. This finding confirms that multi-encoder architecture is more effective than a single-encoder approach for question generation tasks [36].

To further substantiate the role of cognitive level conditioning, we provide empirical evidence contrasting model M9 (without cognitive level conditioning) and the proposed model M10 (with cognitive level conditioning) when generating questions for the same context across all six cognitive levels. Without cognitive level guidance, M9 tends to generate identical questions across all six cognitive levels for the same context, whereas M10 produces clearly differentiated questions that match the intended cognitive demand for each level. Fig. 10 presents a representative example of the difference between the two models using a context about the definition and etymology of history. Without cognitive level guidance, the model generated the same evaluative question for every target cognitive level: ''In your opinion, how effective is the multidimensional approach introduced by Sartono Kartodirdjo? Justify your choice!'' In contrast, the model with cognitive level guidance generated questions that matched the intended cognitive level. For example, it produced a simple definitional question at C1 (''What is the definition of social science according to the text?''), an application question at C3 (''If you were a historian, how would you apply the concept of 'history' to explain the findings of future historical research projects?''), an analysis question at C4 (''Analyze the causal relationship between the Industrial Revolution 4.0 and the emergence of new social problems!''), and a synthesis question at C6 requiring students to create a short story outline connecting historical figures to the origin of a community name. Similar differences were also observed for C2 and C5 but are omitted here for brevity. These results show that cognitive level guidance effectively directs the model to generate questions that align with the intended cognitive level, rather than producing the same type of question for all levels.

Overall, these findings demonstrate that generating cognitively aligned questions requires not only a well-designed architecture, but also a dataset with rich and consistent cognitive level annotations. The proposed model successfully addresses this challenge by combining structural, semantic, and cognitive components into a unified framework, resulting in a model that is capable of generating questions that are

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CONTEXT:
History is the science that studies events, people, countries, or life that occurred in the past. Etymologically, 'history' comes from the Arabic word 'syajaratun' which means 'tree', symbolizing continuous growth and development. ...

QUESTION
M9 (WITHOUT COGNITIVE LEVEL)
In your opinion, how effective is the multidimensional approach introduced by Sartono Kartodirdjo? Justify your choice!

M10 (WITH COGNITIVE LEVEL)
C1 : What is the definition of social science according to the text?
C3 : If you were a historian, how would you apply the concept of 'history' to explain the findings of future historical research projects?
C4 : Analyze the causal relationship between the Industrial Revolution 4.0 and the emergence of new social problems such as 'latent social problems' and 'latent social problems'!
C6 : Create a short story outline about a 7th-century merchant who wants to know the origin of the names 'Jawi Community' and 'Jawi Community'!

FIGURE 10. Example comparison of generated questions for the same context between M9 (without cognitive level conditioning) and M10 (with cognitive level conditioning).

linguistically fluent, contextually relevant, and cognitively appropriate across different levels

B. LIMITATIONS AND GENERALIZABILITY

Despite the promising results, this study has two main limitations. First, the proposed model works best on datasets that are narrative and text-based. A systematic cross-genre analysis (e.g., expository, descriptive, argumentative) could not be conducted within the scope of this study, as both datasets are derived from school textbooks that are not explicitly labeled by text genre. This is evident from the experimental results, where the Social Studies dataset, which consists of explanatory and conceptual texts, produces only 1 keyword overlap across cognitive levels. In contrast, EduProbe, which covers both Social and Natural Science domains, produces up to 3 overlapping keywords, indicating that the model struggles to differentiate cognitive levels when the dataset contains diverse domain content that is not purely narrative. As an indirect indicator of performance variation across differing levels of language complexity, we examine the question length distribution across cognitive levels shown in Fig. 7 Lower cognitive levels (C1-C2) produce shorter questions ($\mu = 8.3$ and $\mu = 11.1$ words, respectively), while higher cognitive levels (C5-C6) produce longer and more elaborate questions ($\mu = 21.5$ and $\mu = 18.8$ words, respectively). This pattern is reflected in representative examples from the Social Studies dataset in Table 7: at C1, ''What is the definition of social science according to the text?'' (11 words) and at C2, ''Explain why Indonesia has abundant and high-quality natural resources!'' (9 words) remain short and directly factual, whereas at C5, ''Critique society's reliance on digital payment systems. What are the potential drawbacks or negative impacts for certain groups in society?'' (20 words) and at C6, ''Make an analogy to explain to your friend why paper money that is not backed by gold still has value?'' (20 words) require substantially longer phrasing to support evaluation and creative reasoning. This illustrates that the length variation observed in Fig. 7 is directly reflected in the actual questions generated by the model, not only in aggregate statistics, suggesting that the model learns language patterns associated with each cognitive level rather than relying solely on keyword matching, even though the datasets do not include explicit text-genre labels.It should also be noted that the lower qualitative scores on EduProbe are not a result of the model failing to generalize, as both models were trained and evaluated independently on their respective datasets. The performance variation is primarily driven by differences in passage length, subject diversity, and keyword variety between the two datasets, rather than domain-specific overfitting. Second, the proposed model still falls short compared to large language models such as GPT-2 used independently, as these models have been trained on significantly larger and more diverse text collections, giving them a stronger baseline language generation capability that is difficult to match with a more constrained architectural approach.

In terms of generalizability, the proposed framework can be deployed in other educational settings provided that two key conditions are met. First, eBloom must be used as cognitive level reference, as the model relies on structured cognitive level annotations to guide the question generation process. Second, the combination of sequential and semantic graph encoders must be maintained, as this architectural choice has been shown to be essential for capturing both the grammatical structure and semantic relationships needed to generate high quality questions. Without these two components, the effectiveness of the proposed framework cannot be guaranteed. It is important to note that the proposed framework is mainly designed for instructional settings where learning objectives and students' cognitive development are key considerations, such as formative assessment, classroom activities, and adap-

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tive learning. We do not claim that the framework is suitable for every assessment context. For example, in high-stakes or selection-based assessments, other factors may be more important than aligning questions with cognitive level.

Cognitive-level guidance is particularly important in contexts where questions must match specific cognitive levels, such as scaffolded instruction for students with different abilities, formative assessments that monitor students' progress from basic recall to higher-order thinking, or exam design that requires a balanced distribution of questions across Bloom's taxonomy. In contrast, a general question generation approach without cognitive-level guidance may be sufficient when the intended cognitive level is not essential, such as for informal comprehension checks, discussion questions that are not linked to a specific learning objective, or quick quizzes where any relevant question is acceptable. This distinction is supported by the ablation results presented in Section V-A. The results show that without cognitive-level guidance, the model tends to generate similar questions instead of adapting to the intended cognitive level, with the generated questions in the cases examined mostly reflecting mid- to higher-order thinking. These findings explain why cognitive-level guidance is especially valuable in instructional and formative assessment contexts, where questions need to match specific learning objectives, while being less critical in contexts where the intended cognitive level is not a primary concern.

Regarding evaluation independence, the qualitative assessment in this study was conducted by three domain expert teachers who were entirely independent of the research team and had no involvement in the development of eBloom or the proposed model. Each teacher independently evaluated the generated questions across three dimensions: relevance, cognitive level, and language quality, without consultation with the other evaluators. Disagreements were resolved through majority voting, resulting in substantial inter-rater agreement (Fleiss' $\kappa = 0.82$). We note that this reflects evaluator-level independence rather than full framework-level independence, as the evaluation rubric itself is derived from the eBloom framework. This evaluation design is nonetheless consistent with established practice in the AQG-for-education literature. Lee et al. [37], in a related study on ChatGPT-based automatic question generation, similarly employed expert panels and in-service teachers who were not involved in the design of the framework to assess question quality through human evaluation. Following this precedent, the present study adopted independent domain-expert teachers as evaluators, consistent with prior methodological practice in the field.

Furthermore, the comparison of inter-rater agreement between standard Bloom's Taxonomy ($\kappa = 0.70$) and eBloom ($\kappa = 0.82$) provides additional evidence that eBloom's enriched cognitive-level definitions reduce ambiguity and promote more consistent judgments across independent raters. While this finding does not fully eliminate the concern regarding circular validation, it suggests that the enhanced conceptual representation in eBloom improves the operational clarity of cognitive-level classification beyond the conventional Bloom's Taxonomy.

VI. CONCLUSION AND FUTURE WORK

Generating questions that are aligned with specific cognitive levels remains a challenging task in educational question generation, as it requires the model to not only understand the input context but also produce questions that reflect the intended depth of thinking. This study addresses this challenge by proposing a cognitive-level guided question generation model that integrates a sequential encoder, a semantic graph encoder, and a cognitive level embedding into a unified framework, with GPT-2 as the decoder to generate fluent and coherent questions. The experimental results demonstrate that the proposed model successfully generates questions that are contextually relevant, cognitively aligned, and grammatically correct across different cognitive levels. The ablation study confirms that each component contributes meaningfully to the overall performance, where the combination of sequential and semantic graph encoders provides both structural and semantic understanding of the input text, while the cognitive level embedding serves as a guiding signal that directs the model to generate questions matched to the intended cognitive level. Furthermore, the keyword overlap analysis reveals that the richness of cognitive level annotation in the training data directly influences the model's ability to differentiate questions across levels, highlighting the importance of dataset quality in cognitive-level question generation.

In terms of future work, several directions can be explored to further improve the proposed framework. First, extending the framework to other domains beyond Social Studies, such as Mathematics, could broaden its applicability and test its robustness across different types of content and cognitive demands. Second, expanding the training dataset with more diverse and domain-specific content could help reduce keyword overlap and improve the model's ability to distinguish between cognitive levels across different domains. Third, exploring a more advanced architectural combination by incorporating large language models such as GPT-2 not only as the decoder but also within the encoder stage, either alongside or as a replacement for the existing sequential and semantic graph encoders, could further enhance the model's ability to capture richer and more nuanced representations of the input text.

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